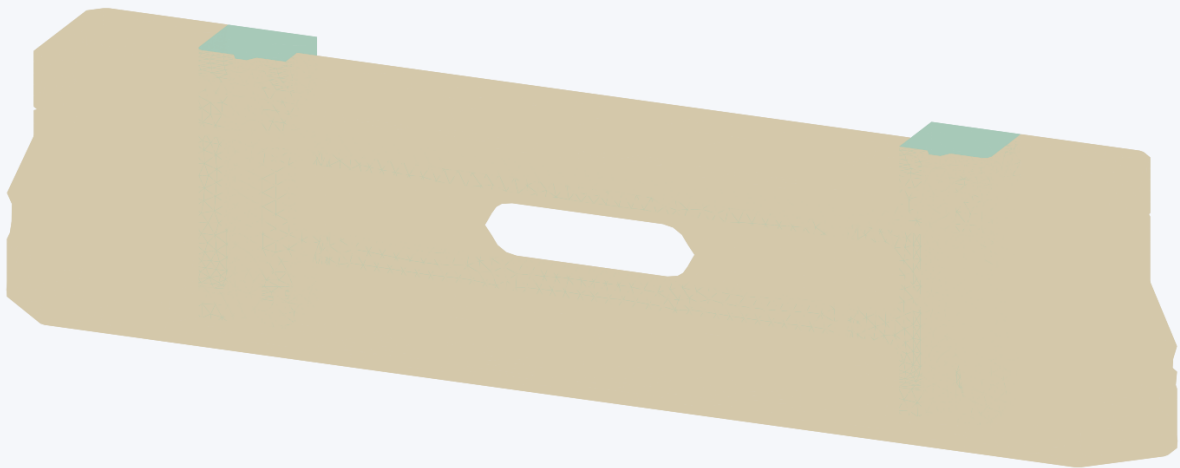


Partial report. Selected-run evidence status: stopped_early

Geometry

Undeformed geometry

Exterior shape from the selected mesh



Mesh-derived exterior • source geometry unchanged

Undeformed geometry from selected mesh. Exterior shape derived from the selected run's mesh; this is not a CAD render.

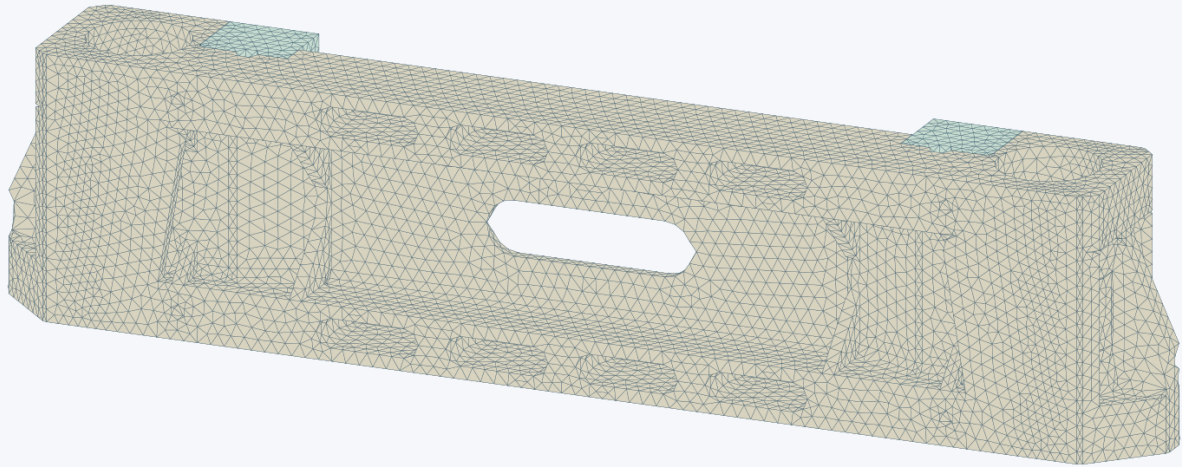
Mesh

Mesh overview and local detail

Mesh nodes	19208	
Tetrahedral elements	67240	
Meshed parts	2	
Nominal element size	4.0	mm
Minimum mean-ratio quality	0.0774	0-1

Mesh overview

Undeformed exterior mesh

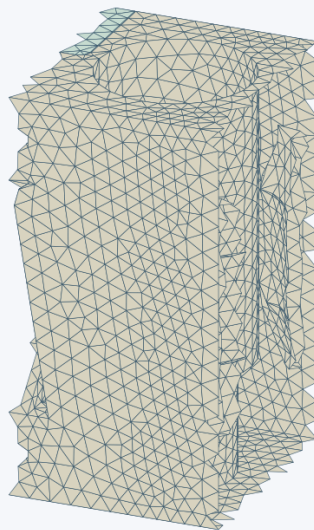


Exterior mesh faces • undeformed geometry

Mesh overview. Undeformed exterior topology of the selected case, verified against its recorded solved-mesh hashes.

Pad-root refinement detail

Pad-root region



Exterior mesh faces • undeformed geometry

Pad-root refinement detail. Positive pad-root region, $X = 112\text{--}150\text{ mm}$. Mesh metadata records a 0.65 target factor (2.6 mm local target).

Mean-ratio quality describes element shape and does not establish stress or time-step convergence.

Setup

Recorded selected-case model

Initial impact velocity	50	m/s
Requested simulation duration	8	ms
Foam depth	203.2	mm
Foam width	41.275	mm
Foam height	152.4	mm
Foam element size	10	mm
Initial contact gap	0.1	mm
Contact friction coefficient	0.1	dimensionless
Foam curve scale	1	dimensionless
Time-step scale	0.6	dimensionless
Requested state-output interval	0.01	ms
Startup mass-scaling DT2MS	-1.5e-07	s
Motor mass	0.1616	kg
Moving mass	0.6503408	kg
Pusher mass	0.4887408	kg
Initial kinetic energy	812.926	J

Recorded material, contact and boundary assumptions

- Body and bolted cable cap flexible; four local perfectly clamped bolt collars. No bolt preload, slip or failure.
- Motor loaded upper-bound inertia distributed over actual bore nodes; no motor housing stiffness or exact inertia tensor.
- PA12-CF10 elastic isotropic proxy $E=1806.6\text{MPa}$ $\nu=.35$ density 1060; print +Z motor axes. No strength/failure claims.
- No fracture, erosion, motor thrust, carrier, released endcap or invented 35g hardware allowance.
- Rigid 8mm aluminum spreaders initially stationary; fixed x,y and rotations, free z without travel limit.
- Pusher transverse bore-quadrant averages are guided by disjoint LINEAR_GLOBAL X/Y equations; every Z DOF is free.

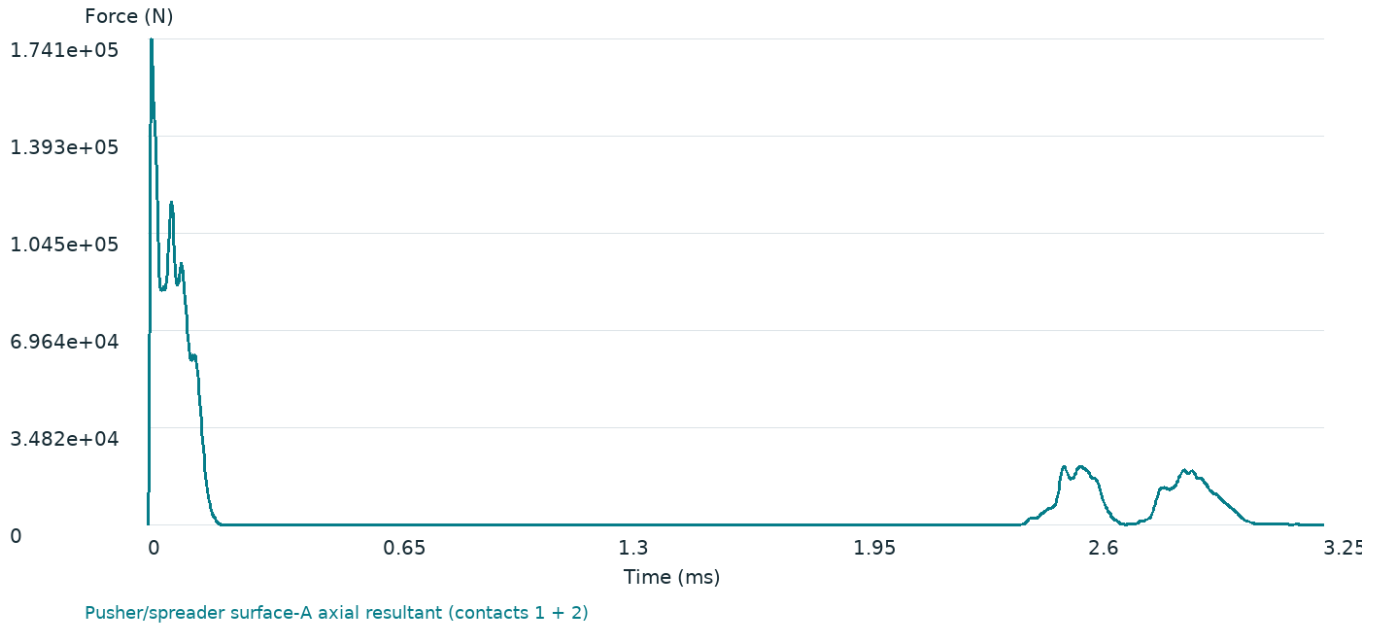
- All axial stopping reaction passes through transient contact; no axial guide constraint.
- Foam rear z fixed, x/y free: frictionless fixed backing; sides unconfined.
- MAT057 synthetic loading and unloading; HU=.05 BETA=0 DAMP=.1 SHAPE=1 FAIL=0. Not a product material model.
- Generic foam anchored at one catalog point; remaining curve chosen for nonbottoming comparative stop.
- Startup-only local mass scaling requested; added mass and inertia error MUST be assessed from solver output.
- Mesh, timestep, energy and momentum require independent assessment.

Solve

Recorded run status	stopped_early	
Recorded reached simulation time	6.16393	ms
Recorded requested simulation time	8	ms
Recorded solver wall time	1138.953	s
Recorded termination state	normal_termination_confirmed	

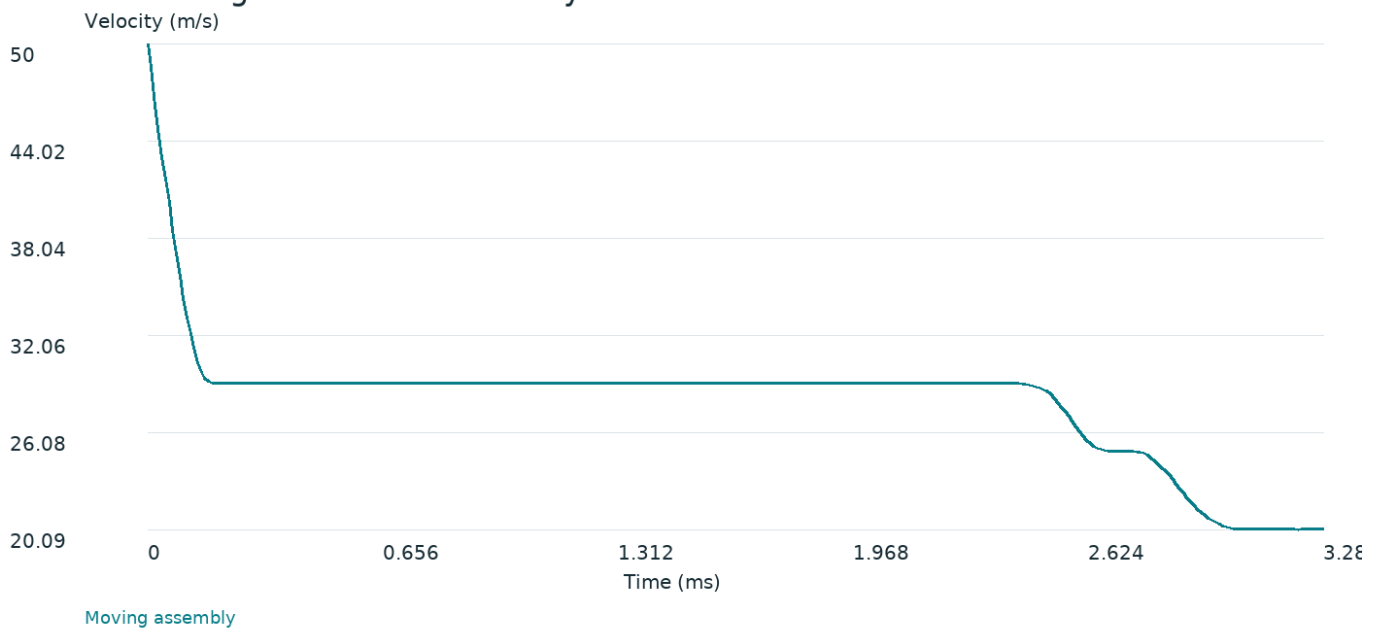
Separate solve charts

Contact force resultant



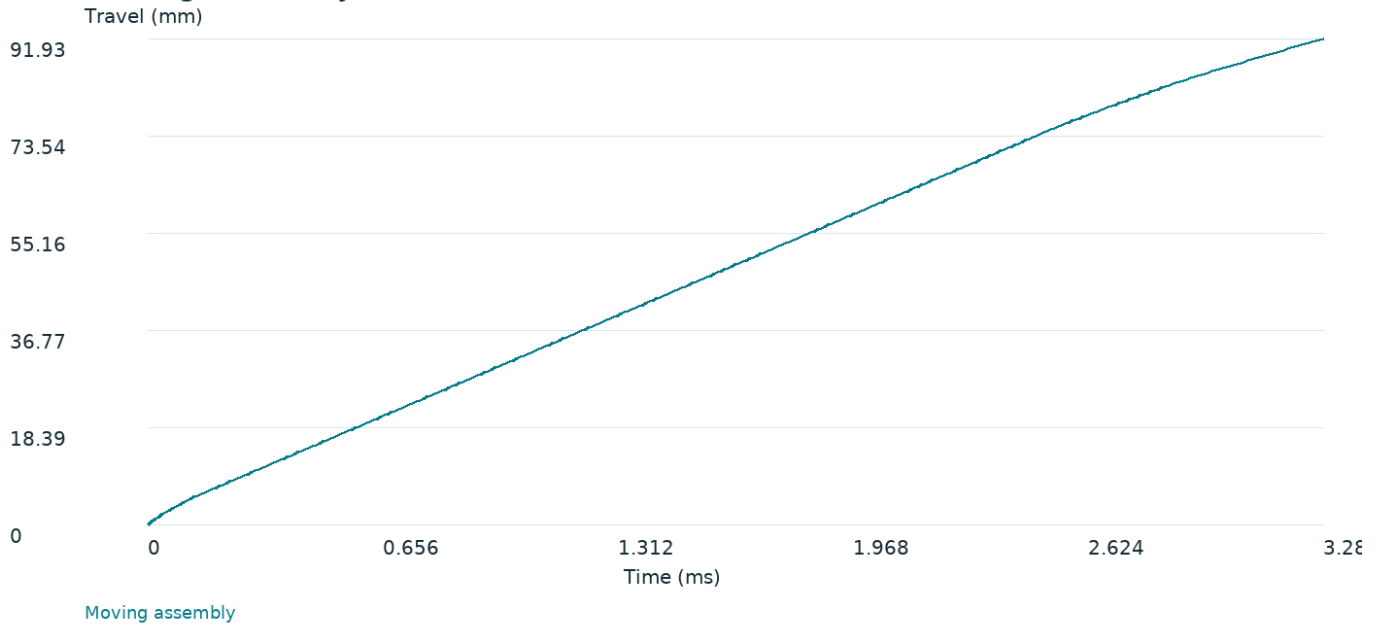
Contact force resultant. projects/compliant_stop_340/cases/guided-4mm-v3/assessment/contact_histories.csv

Mass-weighted forward velocity



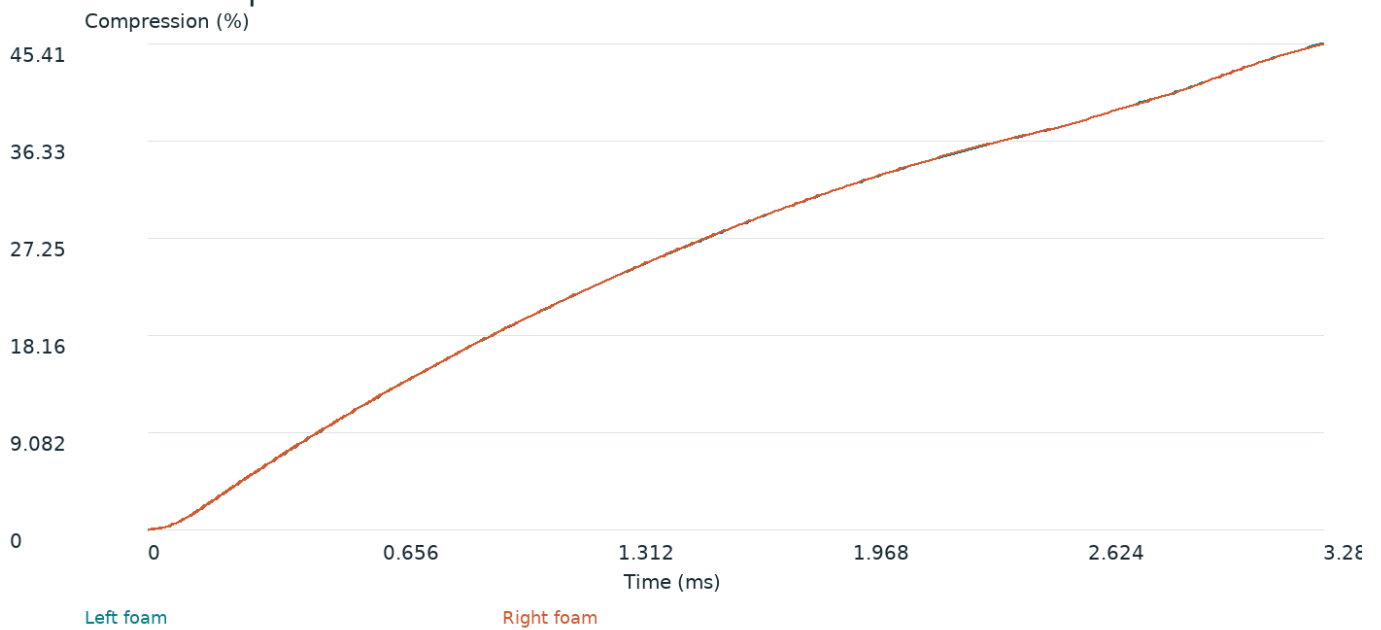
Mass-weighted forward velocity. projects/compliant_stop_340/cases/guided-4mm-v3/assessment/time_histories.csv

Moving assembly travel



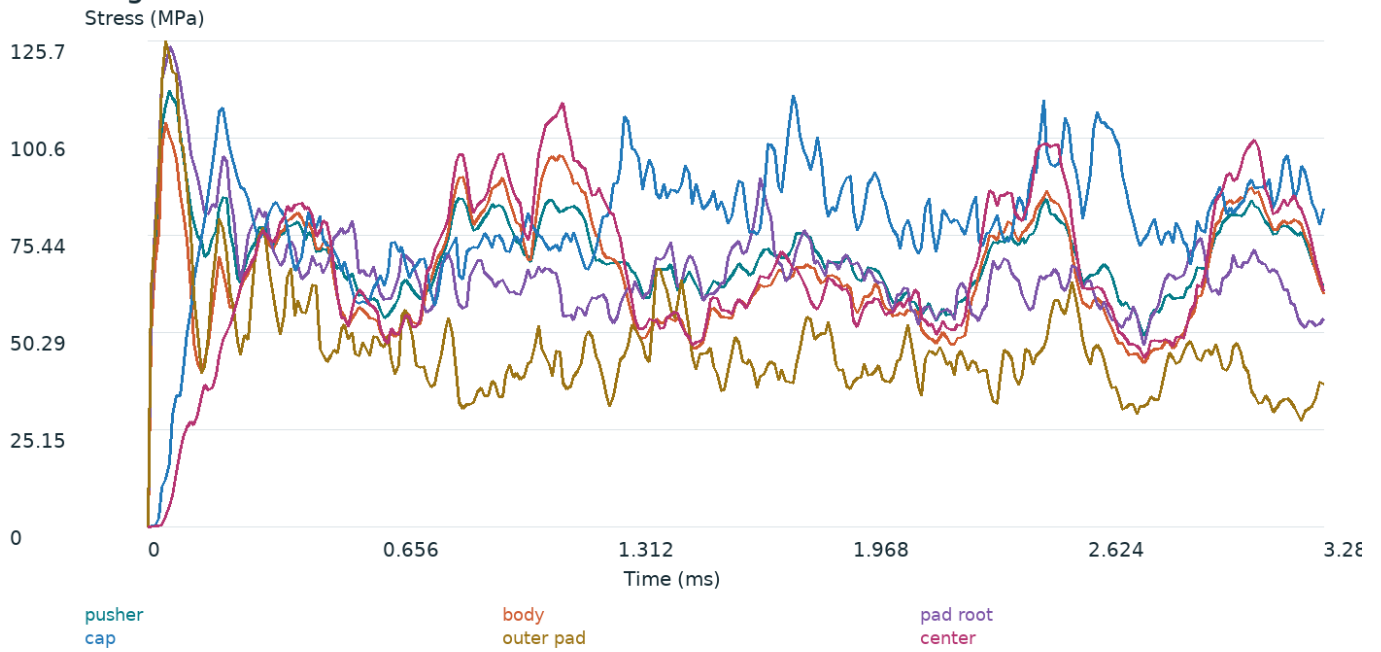
Moving assembly travel. projects/compliant_stop_340/cases/guided-4mm-v3/assessment/time_histories.csv

Foam compression



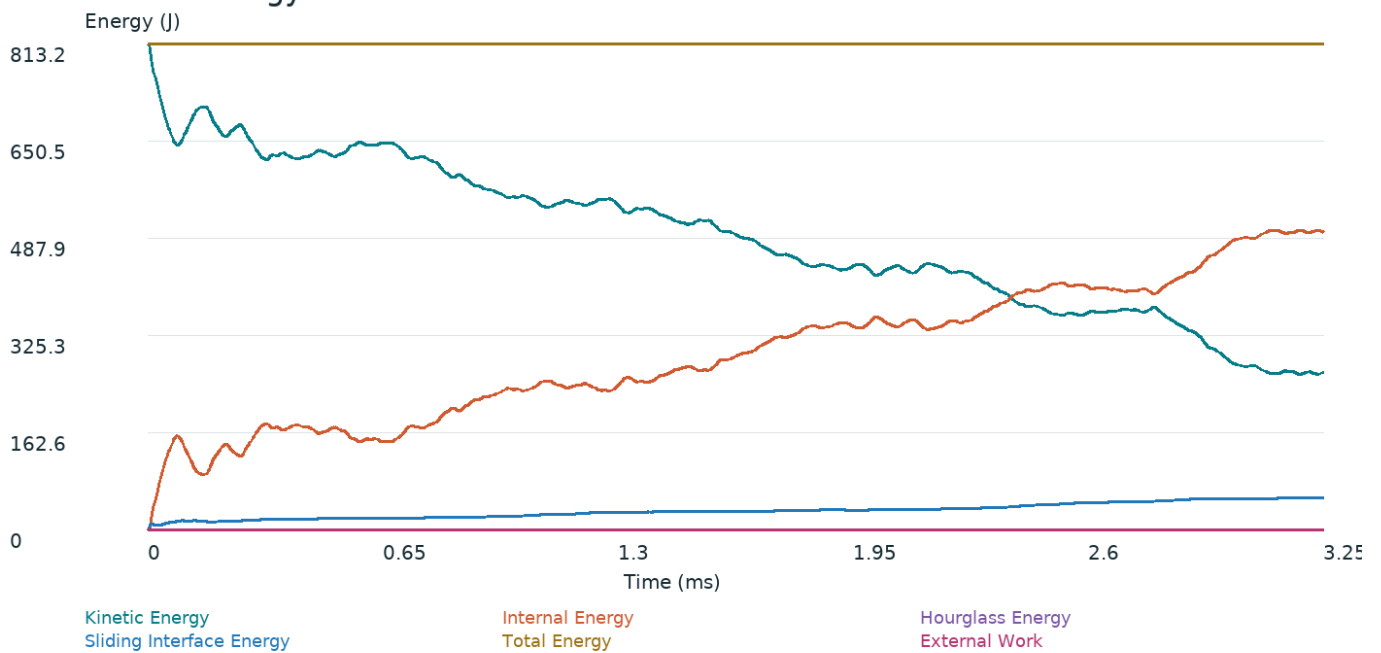
Foam compression. projects/compliant_stop_340/cases/guided-4mm-v3/assessment/time_histories.csv

Regional P99 von Mises stress



Regional P99 von Mises stress. projects/compliant_stop_340/cases/guided-4mm-v3/assessment/stress_histories.csv

Global energy



Global energy. projects/compliant_stop_340/cases/guided-4mm-v3/assessment/global_energy_histories.csv

Results

Run 023 - impact evidence

Evidence status: stopped_early. This report describes only extracted evidence from the selected run. Numerical checks alone do not establish physical qualification.

Geometry and model setup

Recorded model item	Value
kind	impact
pusher kg	0.488741
body kg	0.41625
cable cap kg	0.0724904
motors kg	0.1616
moving kg	0.650341
initial ke J	812.926
spreader each kg	0.135871

Solved mesh

Recorded mesh source: mesh_4mm_repaired. Mesh and timestep convergence are not established by these histories.

Motion and stopping event

Signed forward velocity, travel and foam compression are measured histories. Negative forward velocity denotes rebound; a partial history cannot establish the complete stopping event.

Measured result	Value	Unit
Peak stopping force	174104	N
Maximum travel	91.9267	mm
Peak regional P99 (outer pad)	125.736	MPa
Last extracted time	3.27992	ms
Final forward velocity	20.095	m/s

Regional stress and energy

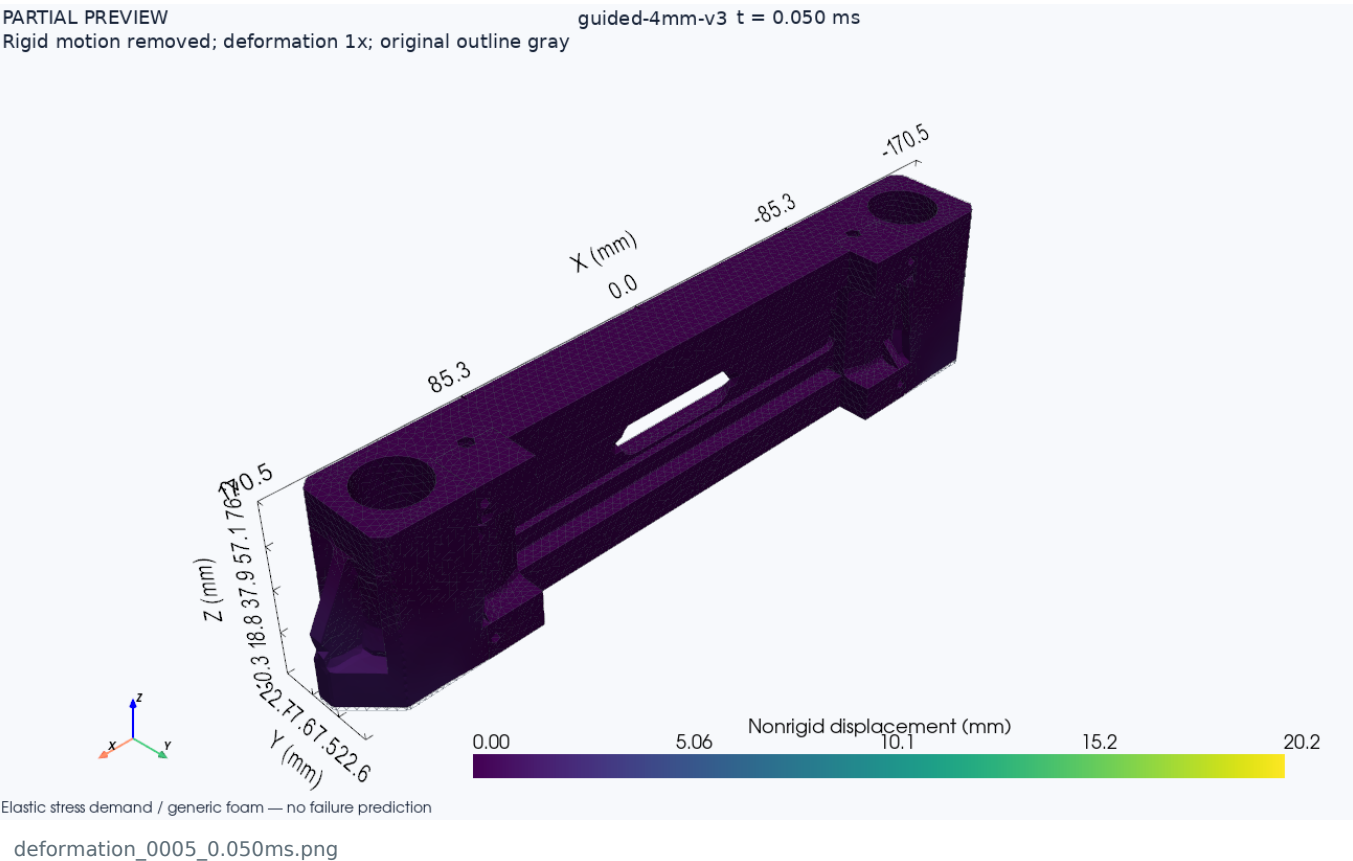
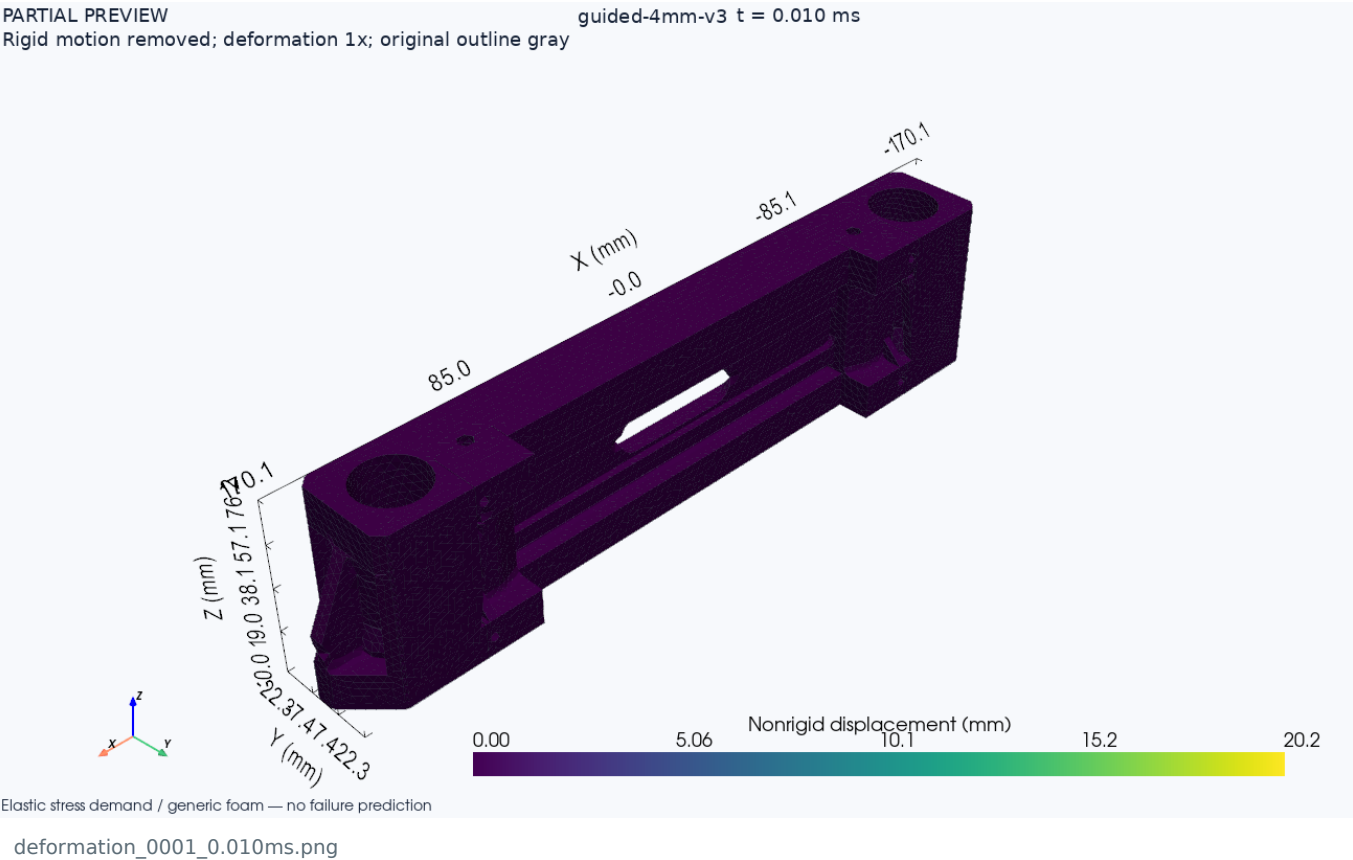
Regional P99 values are elastic demand measures across saved states, not printed-part strength predictions. Force peaks and percentile peaks can occur at different times.

Numerical checks and qualification

Extracted numerical check	Evidence
normal termination	false
requested time reached	false
binary matches nodout	true
moving mass consistent	true
initial energy error below 0p5pct	true
energy balance error below 5pct	true
hourglass below 5pct initial energy	true
sum abs nonfriction contact energy below 5pct initial energy	true
contact impulse momentum error below 5pct	true
no axial spc on moving or guide nodes	true
rigid guides leave only axial translation free	true
no prescribed motion	true
no detected deletion	true
positive sampled foam jacobians	true
first stop captured	false
foam compression below 80pct	true
mass scaling disabled or quantified below 0p1pct moving mass	true
four transverse guide equations satisfied and no axial equation	true

Partial extraction: these histories do not establish a completed solve.

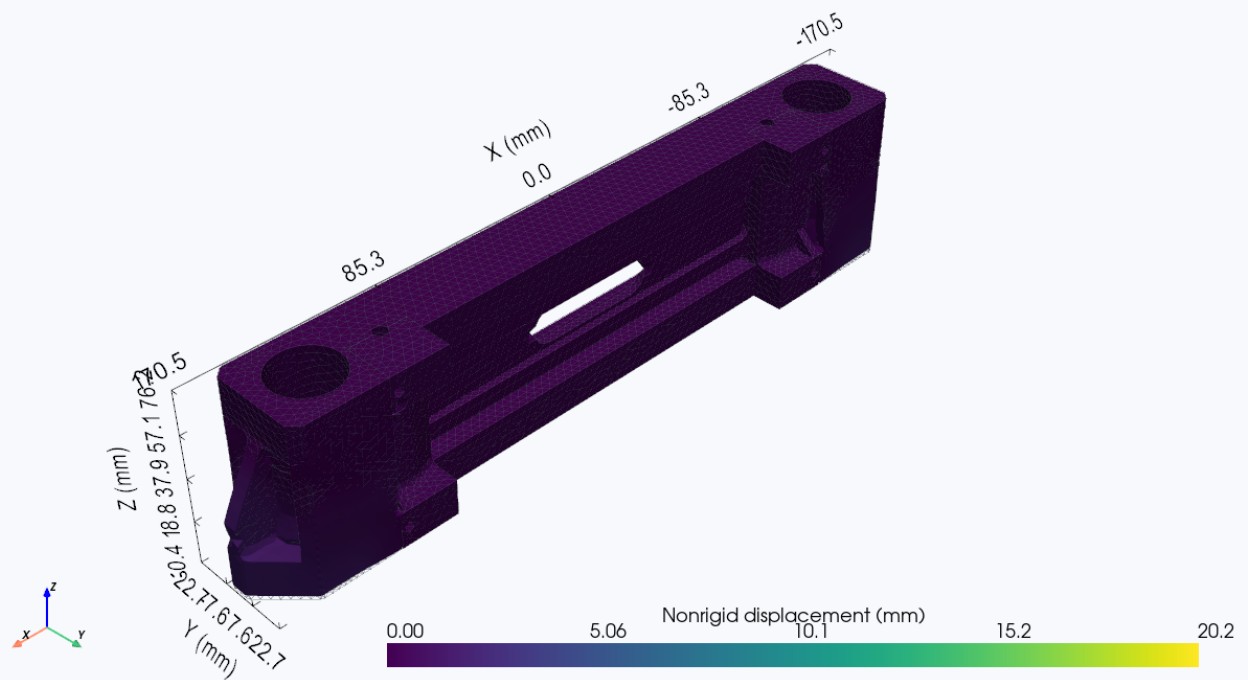
Images, animations and source evidence



PARTIAL PREVIEW

guided-4mm-v3 t = 0.060 ms

Rigid motion removed; deformation 1x; original outline gray



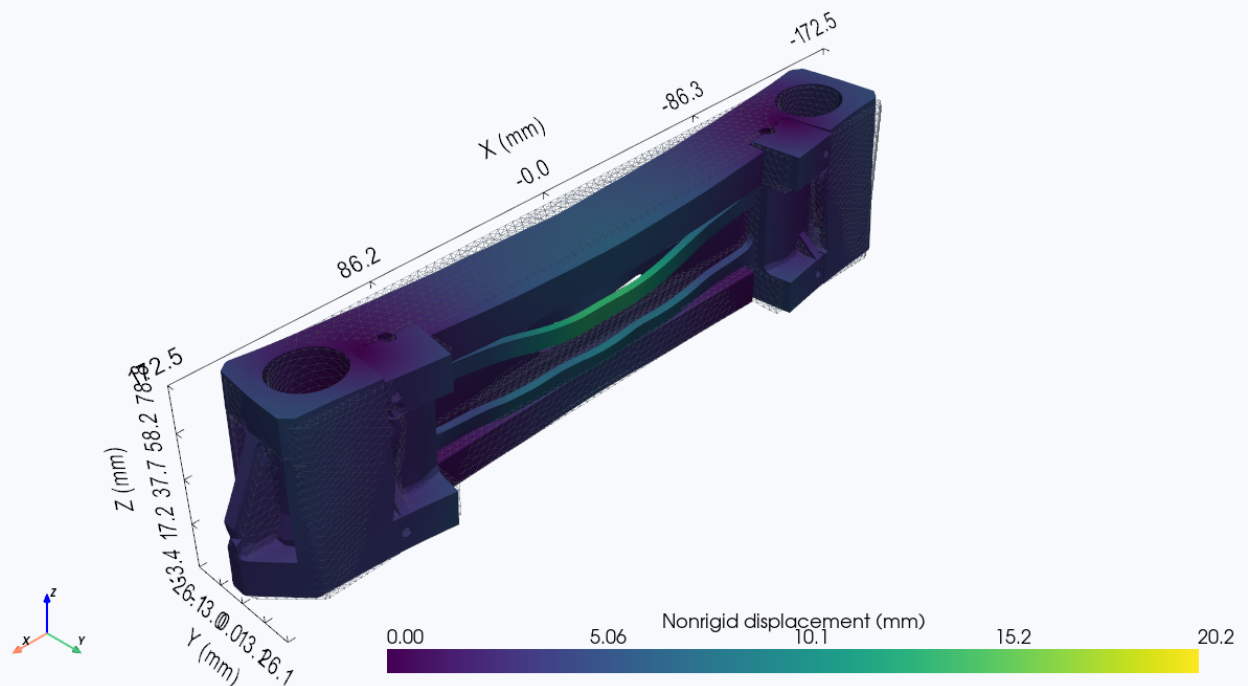
Elastic stress demand / generic foam — no failure prediction

deformation_0006_0.060ms.png

PARTIAL PREVIEW

guided-4mm-v3 t = 1.800 ms

Rigid motion removed; deformation 1x; original outline gray



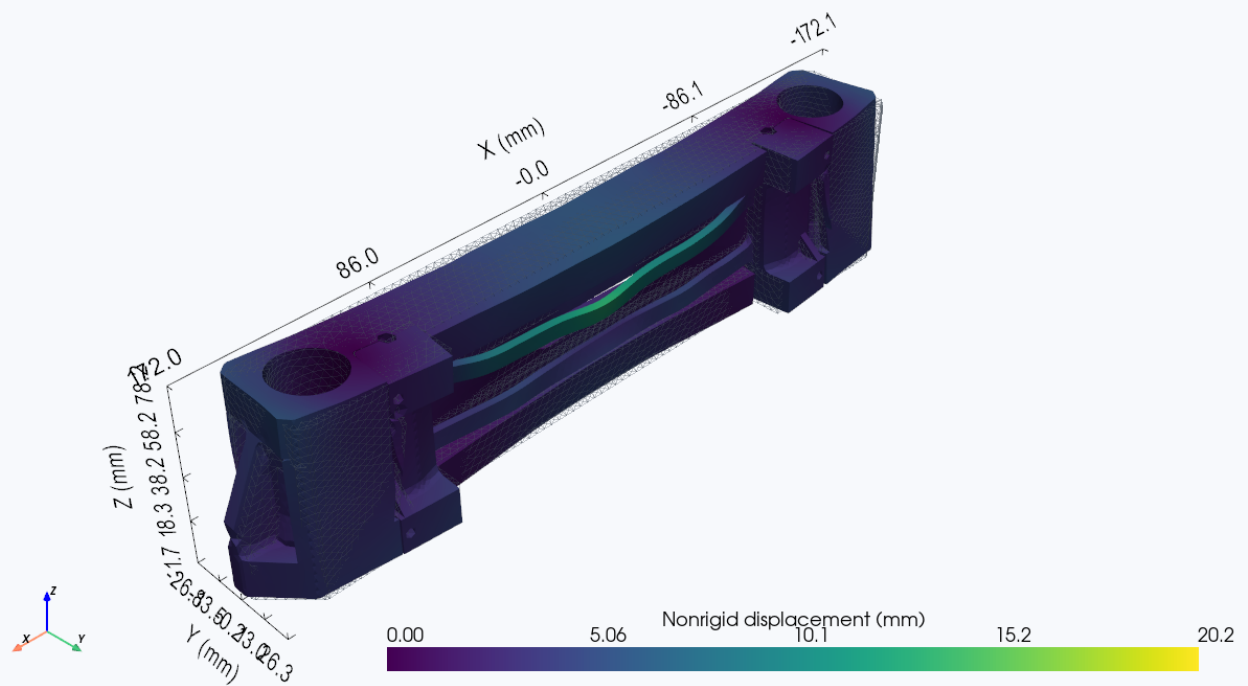
Elastic stress demand / generic foam — no failure prediction

deformation_0180_1.800ms.png

PARTIAL PREVIEW

guided-4mm-v3 t = 1.930 ms

Rigid motion removed; deformation 1x; original outline gray

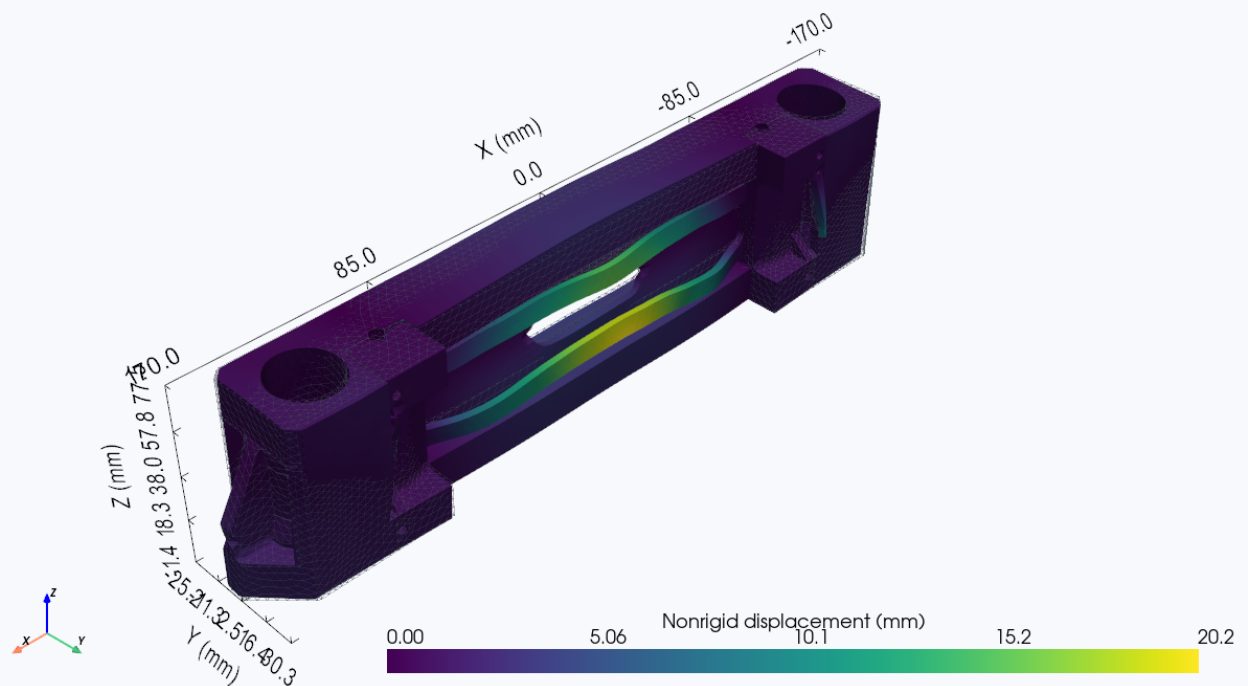


deformation_0193_1.930ms.png

PARTIAL PREVIEW

guided-4mm-v3 t = 2.670 ms

Rigid motion removed; deformation 1x; original outline gray

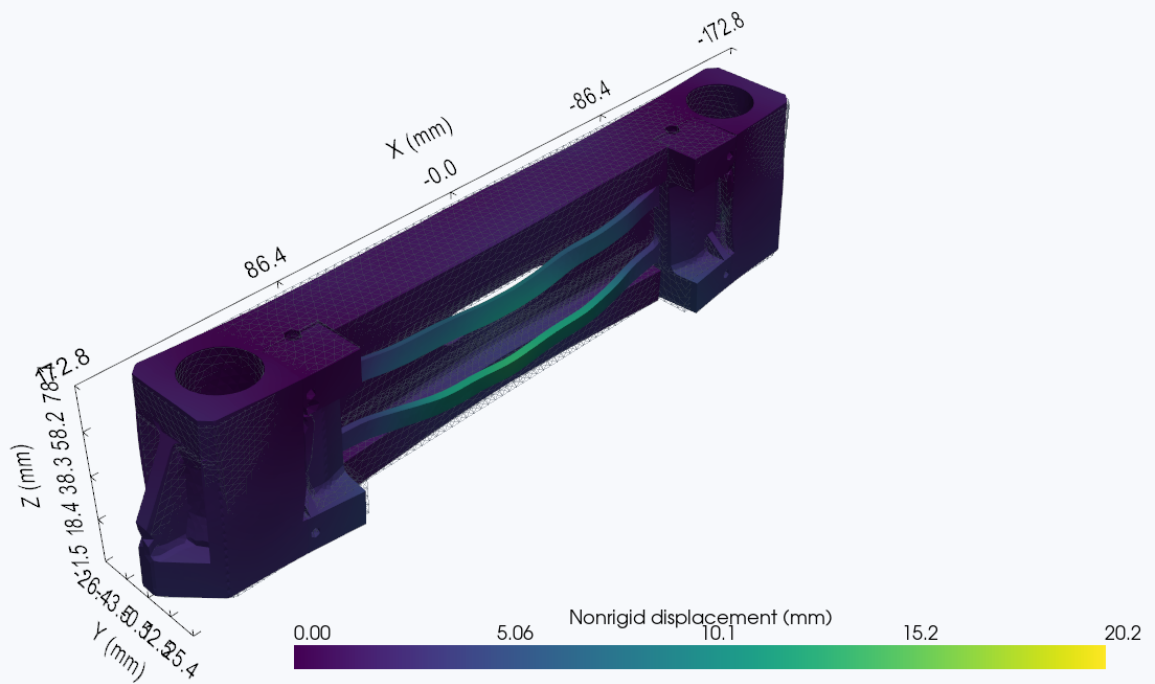


deformation_0267_2.670ms.png

PARTIAL PREVIEW

guided-4mm-v3 t = 3.280 ms

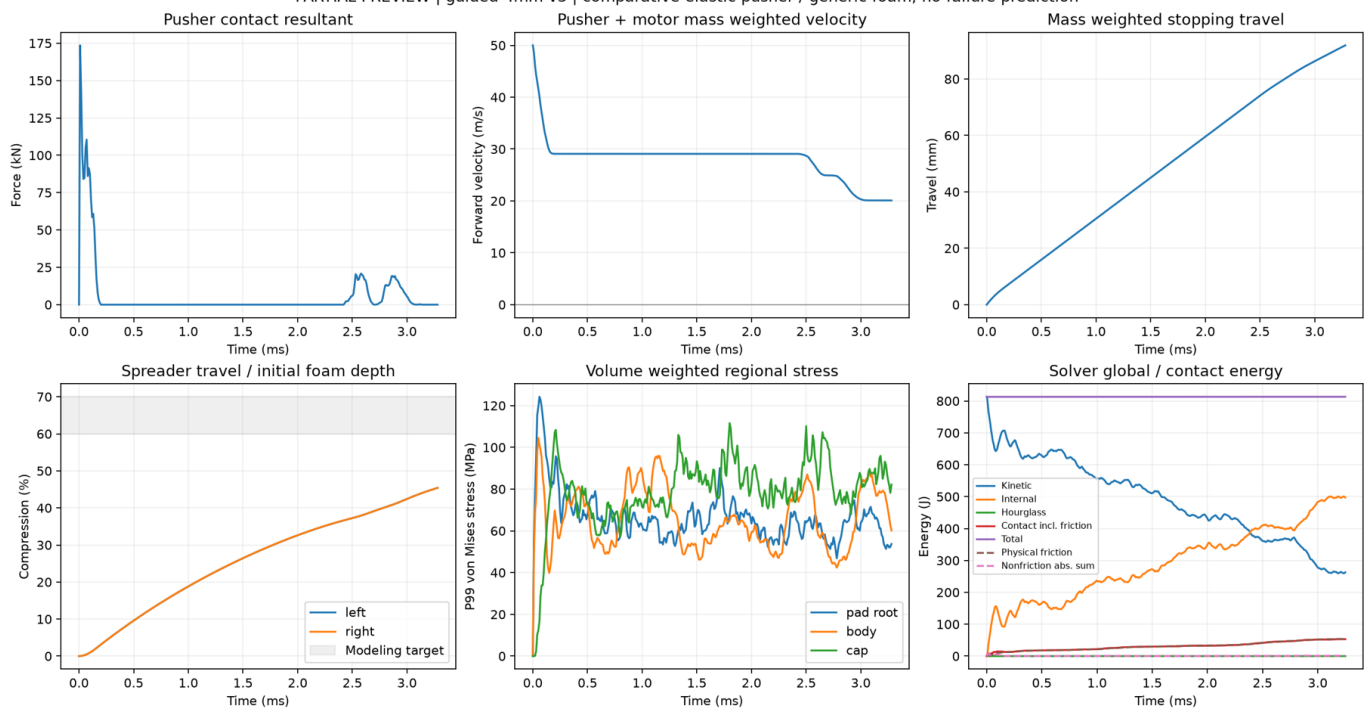
Rigid motion removed; deformation 1x; original outline gray



Elastic stress demand / generic foam — no failure prediction

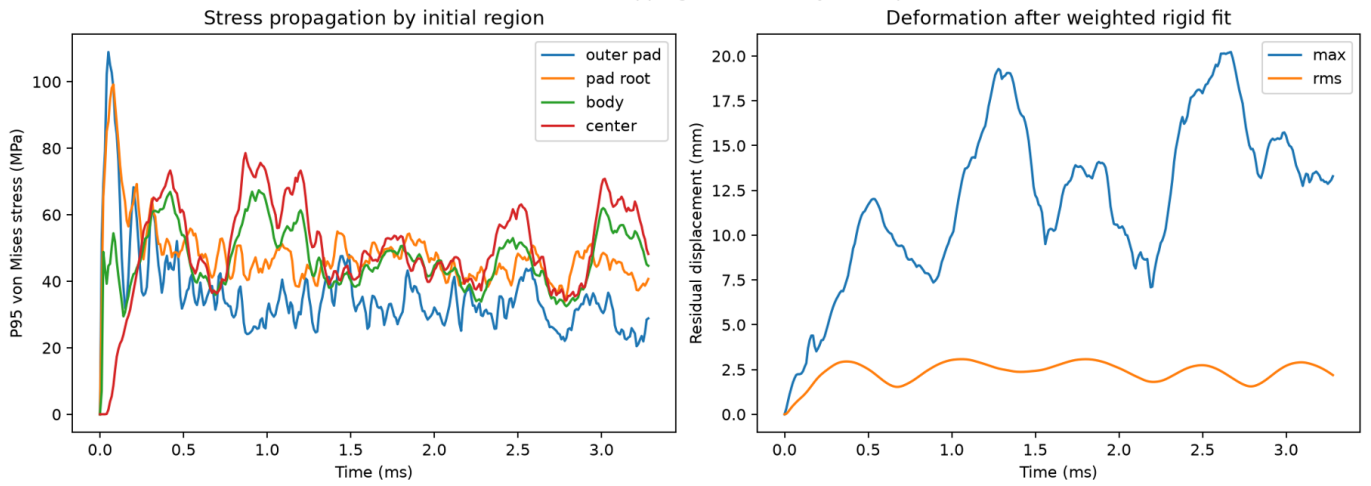
deformation_0328_3.280ms.png

PARTIAL PREVIEW | guided-4mm-v3 | comparative elastic pusher / generic foam; no failure prediction



histories.png

PARTIAL PREVIEW: stopping event is not yet complete

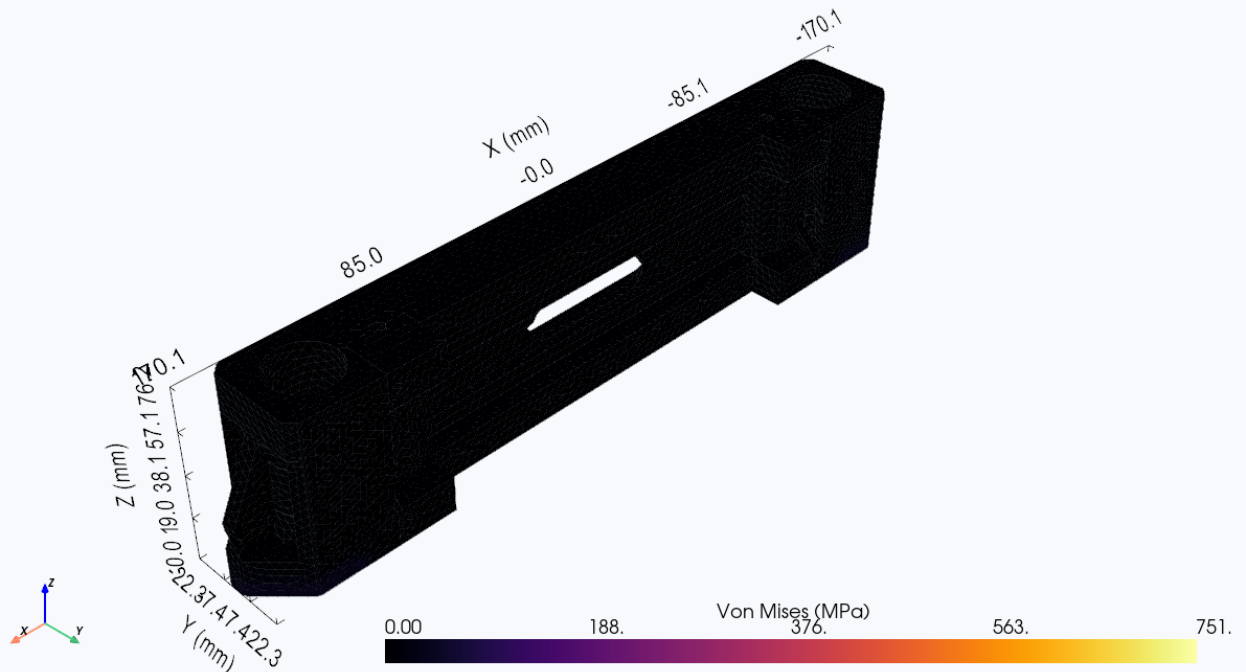


stress-propagation-deformation.png

PARTIAL PREVIEW

guided-4mm-v3 t = 0.010 ms

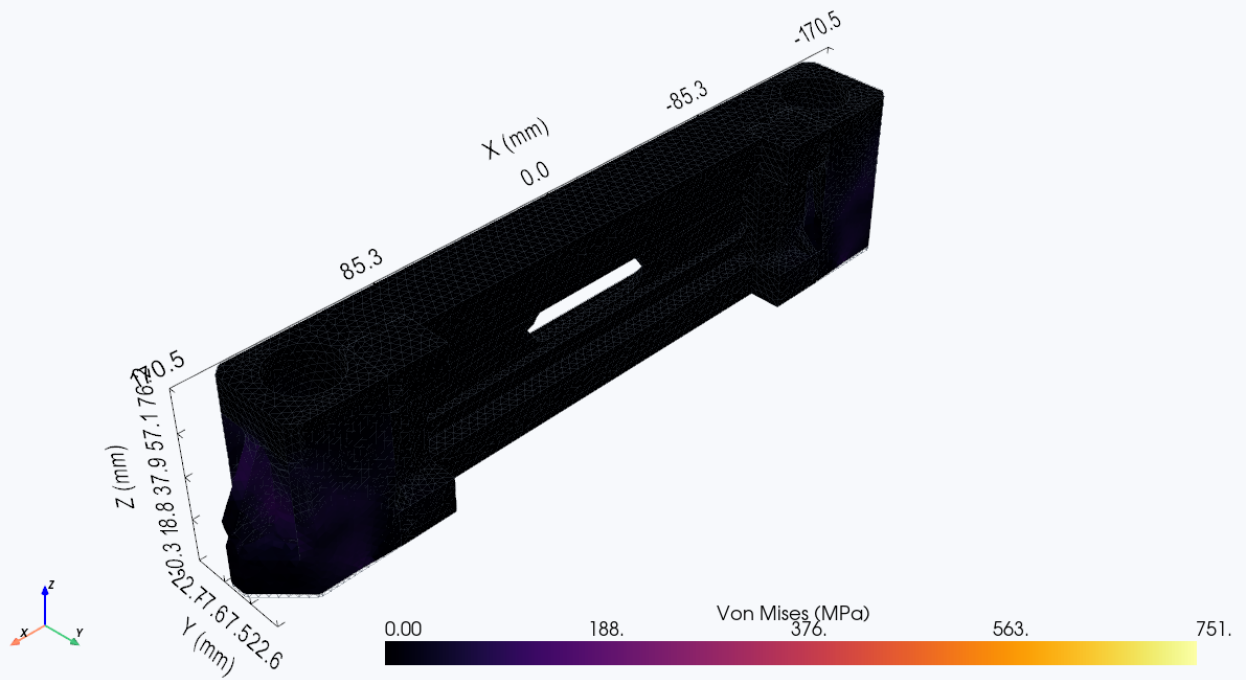
Rigid motion removed; deformation 1x; original outline gray



Elastic stress demand / generic foam — no failure prediction

stress_0001_0.010ms.png

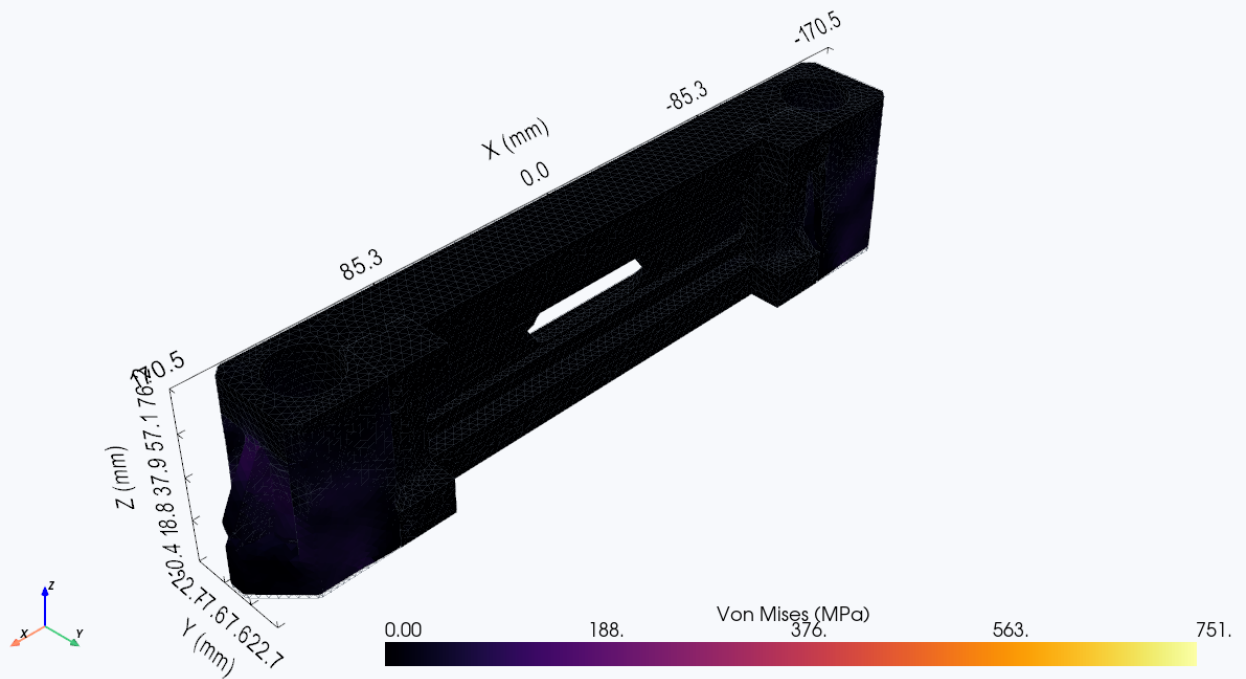
PARTIAL PREVIEW
guided-4mm-v3 t = 0.050 ms
Rigid motion removed; deformation 1x; original outline gray



Elastic stress demand / generic foam — no failure prediction

stress_0005_0.050ms.png

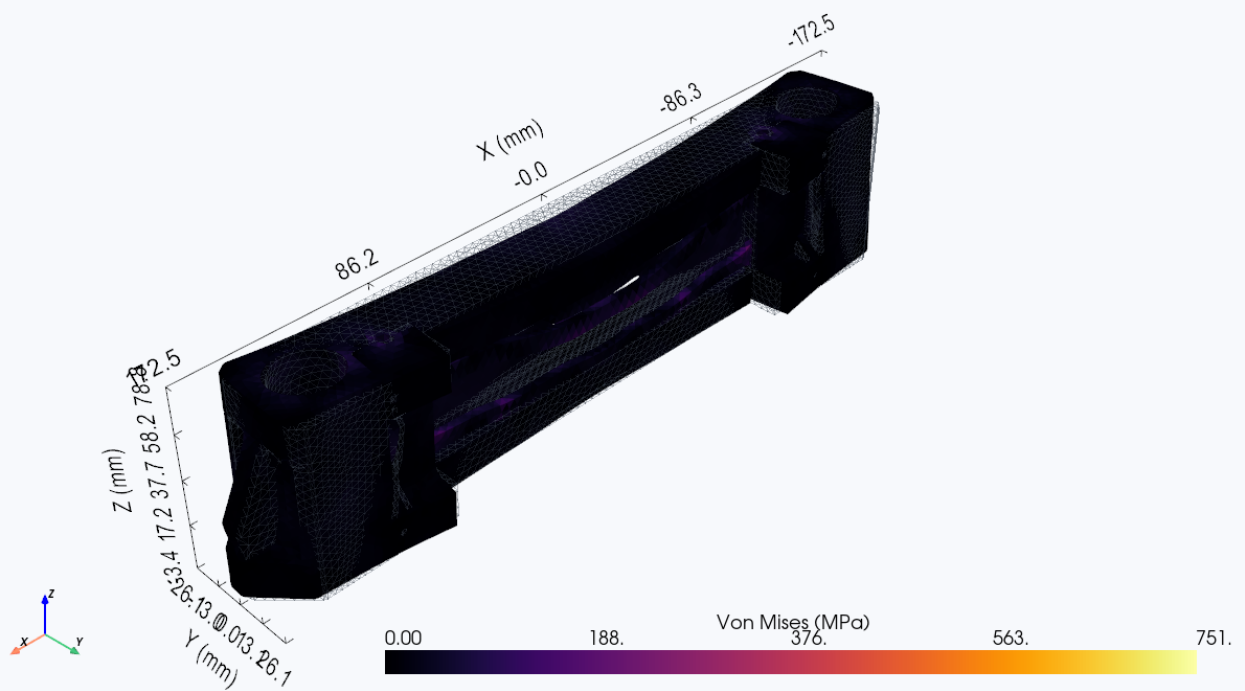
PARTIAL PREVIEW
guided-4mm-v3 t = 0.060 ms
Rigid motion removed; deformation 1x; original outline gray



Elastic stress demand / generic foam — no failure prediction

stress_0006_0.060ms.png

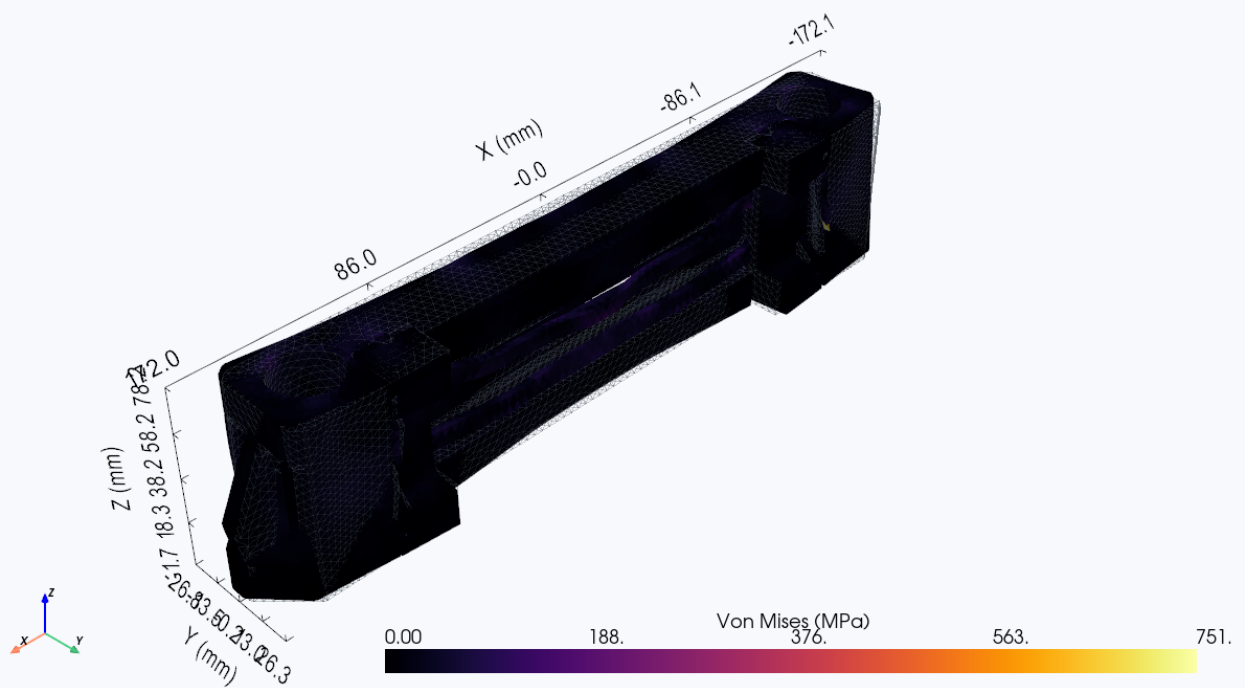
PARTIAL PREVIEW guided-4mm-v3 t = 1.800 ms
 Rigid motion removed; deformation 1x; original outline gray



Elastic stress demand / generic foam — no failure prediction

stress_0180_1.800ms.png

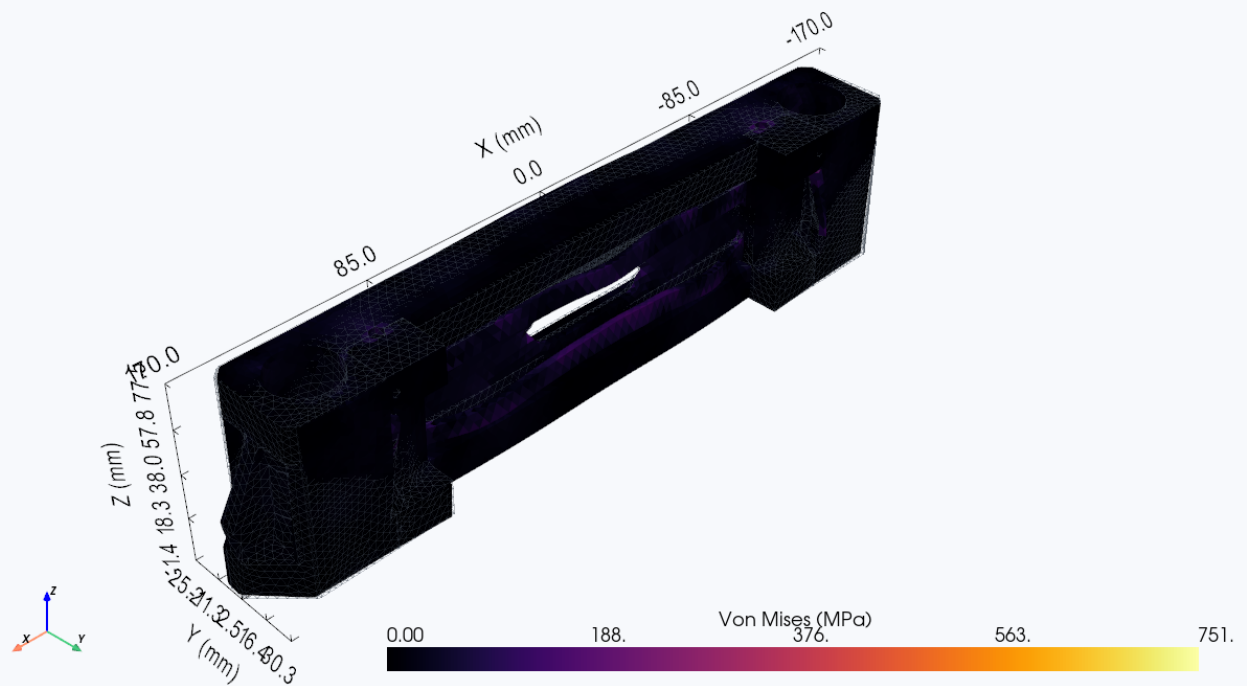
PARTIAL PREVIEW guided-4mm-v3 t = 1.930 ms
 Rigid motion removed; deformation 1x; original outline gray



Elastic stress demand / generic foam — no failure prediction

stress_0193_1.930ms.png

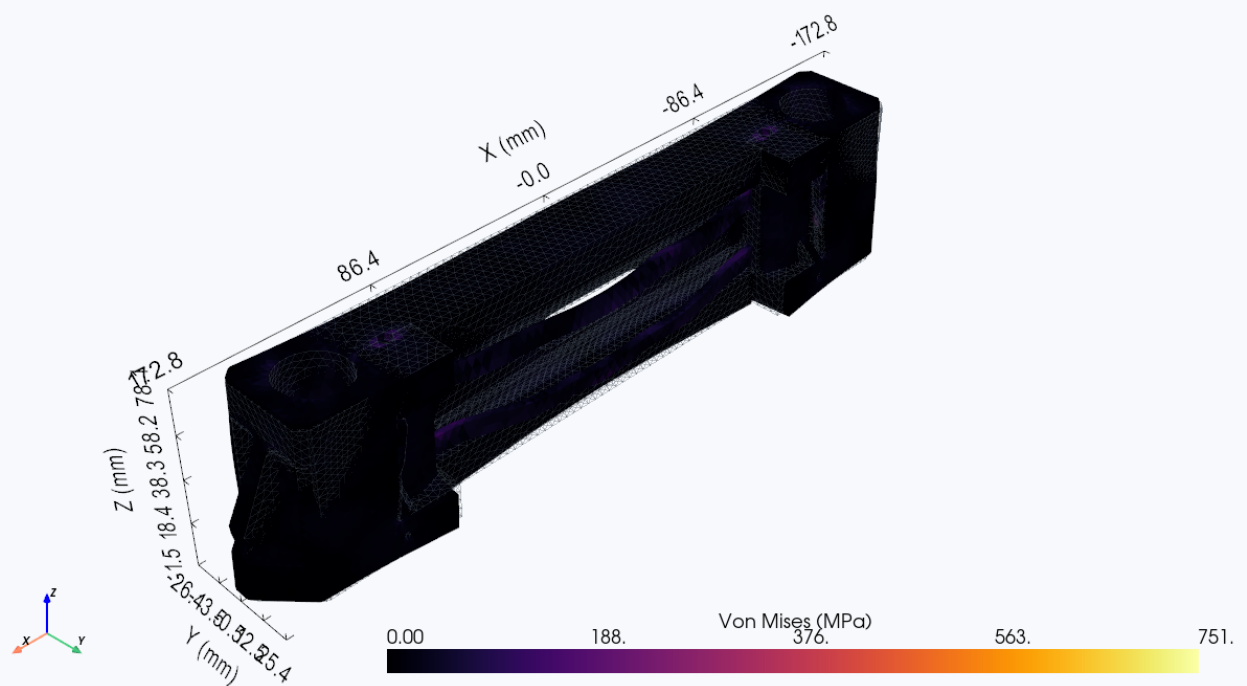
PARTIAL PREVIEW guided-4mm-v3 t = 2.670 ms
 Rigid motion removed; deformation 1x; original outline gray



Elastic stress demand / generic foam — no failure prediction

stress_0267_2.670ms.png

PARTIAL PREVIEW guided-4mm-v3 t = 3.280 ms
 Rigid motion removed; deformation 1x; original outline gray



Elastic stress demand / generic foam — no failure prediction

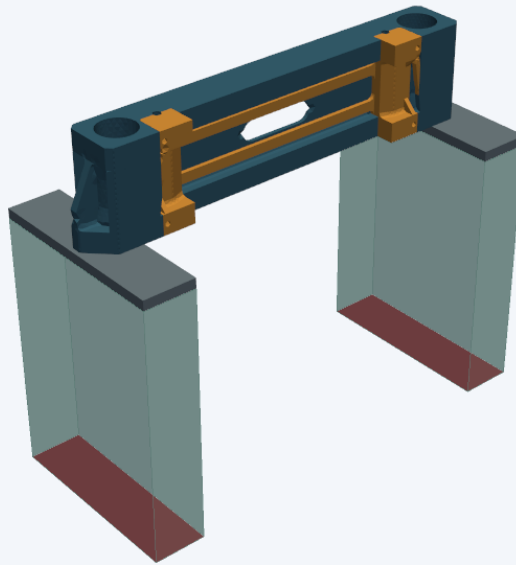
stress_0328_3.280ms.png

PARTIAL PREVIEW

WORLD MOTION guided-4mm-v3

Time 0.000 ms; impact along -Z

- ▲ Generic foam
- ▲ Fixed backing
- ▲ Aluminum plates
- ▲ Pusher body
- ▲ Cable cap



Pusher forward speed 50.00 m/s travel -0.0 mm

Foam compression L/R -0.0% / -0.0%

True displacement 1x

component colors, not a stress contour

Foam initial outline in gray; axes and geometry in mm

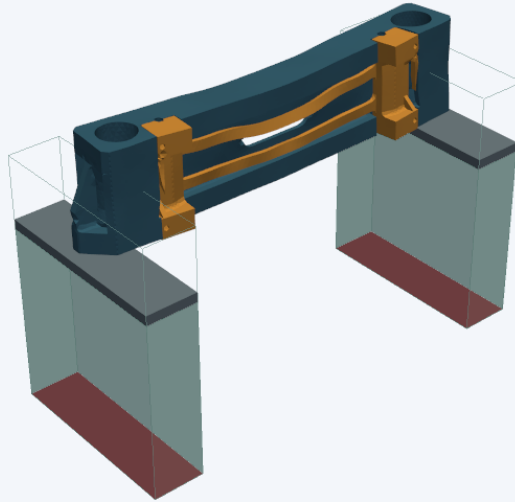
world_0000_0.000ms.png

PARTIAL PREVIEW

WORLD MOTION guided-4mm-v3

Time 1.690 ms; impact along -Z

- ▲ Generic foam
- ▲ Fixed backing
- ▲ Aluminum plates
- ▲ Pusher body
- ▲ Cable cap



Pusher forward speed 29.08 m/s travel 50.5 mm

Foam compression L/R 28.9% / 29.0%

True displacement 1x

component colors, not a stress contour

Foam initial outline in gray; axes and geometry in mm

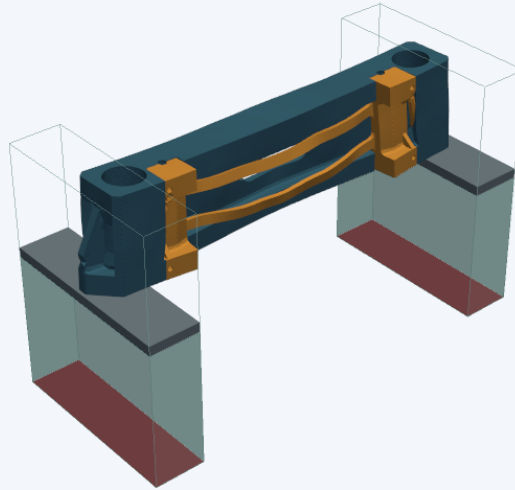
world_0169_1.690ms.png

PARTIAL PREVIEW

Time 3.280 ms; impact along -Z

WORLD MOTION guided-4mm-v3

- ▲ Generic foam
- ▲ Fixed backing
- ▲ Aluminum plates
- ▲ Pusher body
- ▲ Cable cap



Pusher forward speed 20.10 m/s travel 91.9 mm

Foam compression L/R 45.4% / 45.4%

True displacement 1x

component colors, not a stress contour

Foam initial outline in gray; axes and geometry in mm

world_0328_3.280ms.png

- cable_guide_motion.csv
- contact_energy_by_interface.csv
- contact_energy_components.csv
- contact_histories.csv
- front_bore_clearance_diagnostic.json
- global_energy_histories.csv
- metrics.json
- part_energy_histories.csv
- sampled_planar_penetration.csv
- spc_histories.csv
- stress_histories.csv
- time_histories.csv
- weighted_guide_constraints.csv

Generated from selected-run extracted evidence; no authored report is available. Animation figures and their captions are excluded from this PDF; animations remain available in the standalone HTML.